2.38 Establish each of the following equivalences, where $\phi, \psi, \theta$, and all the $\theta_i$ are formulas.

(a) $((\phi \wedge \psi) \vee \neg\theta) \equiv ((\phi \vee \neg\theta) \wedge (\psi \vee \neg\theta))$
(b) $(\phi \to \neg\psi) \equiv (\psi \to \neg\phi)$
(c) $(\theta \to (\phi \vee \psi)) \equiv (\phi \vee (\psi \vee \neg\theta))$
(d) $((\theta_1 \wedge \theta_2) \wedge (\theta_3 \wedge \theta_4)) \equiv (\theta_1 \wedge ((\theta_2 \wedge \theta_3)) \wedge \theta_4))$

*Proof of (a).*

$$\begin{aligned} ((\phi \wedge \psi) \vee \neg\theta) &\equiv (\neg\theta \vee (\phi \wedge \psi)) && \text{(by Theorem 2.3(b))} \\ &\equiv ((\neg\theta \vee \phi) \wedge (\neg\theta \vee \psi)) && \text{(by Theorem 2.3(k) and transitivity)} \end{aligned}$$

But
$(\neg\theta \vee \phi) \equiv (\phi \vee \neg\theta)$ and $(\neg\theta \vee \psi \equiv (\psi \vee \neg\theta)$
by Theorem 2.3(b) so Exercise 2.37(b)
$((\neg\theta \vee \phi) \wedge (\neg\theta \vee \psi)) \equiv ((\phi \vee \neg\theta) \wedge (\psi \vee \neg\theta))$.
Then by transitivity of logical equivalence,
$((\phi \wedge \psi) \vee \neg\theta) \equiv ((\phi \vee \neg\theta) \wedge (\psi \vee \neg\theta))$

□

*Proof of (b).* One of the logical equivalences Theorem 2.4(a) states is: $(\phi \to \psi) \equiv (\neg\psi \to \neg\phi)$.

Treating the formulas above as metavariables and substituting $\phi \mapsto \phi, \psi \mapsto \neg\psi$ uniformly on both sides we arrive at $(\phi \to \neg\psi) \equiv (\neg\neg\psi \to \neg\phi)$.

Theorem 2.3 (g) states the law of double negation which is $\neg\neg\phi \equiv \phi$. Substituting $\phi \mapsto \psi$ uniformly on both sides we arrive at $\neg\neg\psi \equiv \psi$.

So then by exercise 2.37(d) $(\neg\neg\psi \to \neg\phi) \equiv (\psi \to \neg\phi)$ and transitively we get $(\phi \to \neg\psi) \equiv (\psi \to \neg\phi)$. □

*Proof of (c).* One of the logical equivalences Theorem 2.4(a) states is: $(\phi \to \psi) \equiv (\neg\phi \vee \psi)$.

Treating the above formulas as metavariables and substituting $\phi \mapsto \theta, \psi \mapsto (\phi \vee \psi)$ uniformly on both sides we arrive at $(\theta \to (\phi \vee \psi)) \equiv (\neg\theta \vee (\phi \vee \psi))$.

By Theorem 2.3 (b) we have commutativity of $\vee$ so, $(\neg\theta \vee (\phi \vee \psi)) \equiv ((\phi \vee \psi) \vee \neg\theta)$.

By Theorem 2.3 (f) we have associativity of $\vee$, so together with symmetry, $((\phi \vee \psi) \vee \neg\theta) \equiv (\phi \vee (\psi \vee \neg\theta))$.

Lastly by transitivity we arrive at $(\theta \to (\phi \vee \psi)) \equiv (\phi \vee (\psi \vee \neg\theta))$. □

*Proof of (d).* By Theorem 2.3 (e) we have $(\phi \wedge (\psi \wedge \theta)) \equiv ((\phi \wedge \psi) \wedge \theta)$.

Then by symmetry we have $((\phi \wedge \psi) \wedge \theta) \equiv (\phi \wedge (\psi \wedge \theta))$.

Treating the above formulas as metavariables and substituting $\phi \mapsto \theta_1, \psi \mapsto \theta_2, \theta \mapsto (\theta_3 \wedge \theta_4)$ uniformly on both sides we get $((\theta_1 \wedge \theta_2) \wedge (\theta_3 \wedge \theta_4)) \equiv (\theta_1 \wedge (\theta_2 \wedge (\theta_3 \wedge \theta_4)))$.

By Theorem 2.3 (e) once again, substituting $\phi \mapsto \theta_2, \psi \mapsto \theta_3, \theta \mapsto \theta_4$ uniformly on both sides we get $(\theta_2 \wedge (\theta_3 \wedge \theta_4)) \equiv ((\theta_2 \wedge \theta_3) \wedge \theta_4)$. Then by Exercise 2.37(b) (with $\theta_1 \equiv \theta_1$, from Exercise 2.35(a)), $(\theta_1 \wedge (\theta_2 \wedge (\theta_3 \wedge \theta_4))) \equiv (\theta_1 \wedge ((\theta_2 \wedge \theta_3) \wedge \theta_4))$.

Lastly by transitivity we arrive at $((\theta_1 \wedge \theta_2) \wedge (\theta_3 \wedge \theta_4)) \equiv (\theta_1 \wedge ((\theta_2 \wedge \theta_3) \wedge \theta_4))$. $\square$